

SAVITRIBAI PHULE PUNE UNIVERSITY



A FINAL PROJECT REPORT

ON

TITLE OF THE PROJECT

SUBMITTED TOWARDS THE
PARTIAL FULFILLMENT OF THE REQUIREMENTS OF

BACHELOR OF ENGINEERING

BY

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Under The Guidance of

Guide Name



DEPARTMENT OF COMPUTER ENGINEERING
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A.Y. 2025–2026



Pimpri Chinchwad College Of Engineering And Research
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CERTIFICATE

This is to certify that the Project Entitled

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is a bonafide work carried out by Students under the supervision of **Prof. Guide Name** and it is submitted towards the partial fulfillment of the requirement of **Bachelor of Engineering** (Computer Engineering) Project.

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Dept. of Computer Engg.

Name of Project Coordinator
Project Co-ordinator

Prof.Dr. H.U.Tiwari
Principal

EXTERNAL SIGNATURE

Acknowledgments

It gives us great pleasure in presenting the preliminary project report on '**Title of the Project**'.

I would like to take this opportunity to thank my internal guide **Prof. Guide Name** for giving me all the help and guidance needed. I am really grateful to them for their kind support. Their valuable suggestions were very helpful. I would also like to thank our Project Coordinator **Prof. Project Coordinator Name** for their guidance and support.

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In the end our special thanks to **Other Person Name** for providing various resources such as laboratory with all needed software platforms, continuous Internet connection, for our Project.

Name of student 1
Name of student 2
Name of student 3
Name of student 4
(B.E. Computer Engg.)

Abstract

Enter a maximum 250 words abstract that defines the idea of your project. Include introduction, challenges faced and solution using the methodology should be included in the abstract.

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CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

Explain in brief overview of your project.

1.2 NEED AND RELEVANCE

Explain in brief Need and Relevance of your project.

1.3 MOTIVATION

Explain in brief motivation of your project.

1.4 PROBLEM STATEMENT

Explain Problem statement in one line.

1.5 AIM AND OBJECTIVES

Explain Aim. Explain objectives (In points). **Objectives:**

- Objective 1
- Objective 2
- Objective 3
- Objective 4
- Objective 5

1.6 PROJECT SCOPE AND LIMITATIONS

Explain scope of the project.

Explain limitations of the project.

1.7 ORGANISATION OF THE REPORT

Explain the structure of Report.

CHAPTER 2

LITERATURE SURVEY

2.1 OVERVIEW OF EXISTING RESEARCH

Short overview of existing research in words.

2.2 LITERATURE SURVEY

Table 2.1: Literature Survey

Sr. No	Paper Title	Journal Name	Authors & Publication Date	Methodology	Research Gap
1	Paper Title 1	Journal 1	Author1, 2022	Method used	Gap identified
2	Paper Title 2	Journal 2	Author2, 2023	Method used	Gap identified
3	Paper Title 3	Journal 3	Author3, 2024	Method used	Gap identified

2.3 RESEARCH GAP ANALYSIS

Explain the challenges in brief.

CHAPTER 3

SOFTWARE REQUIREMENT

SPECIFICATION

3.1 INTRODUCTION

3.1.1 Purpose and Scope of Document

1. Facilitating other Documentation: The SRS forms the basis for a load of other important documents such as the Software Design Specification.
2. Product Validation: It basically helps in validating with the client that the product which is being delivered, meets what they asked for.

3.1.2 Characteristics of a Software Requirement Specification

1. Accuracy: This is the first and foremost requirement. The development team will get nowhere if the SRS which will be the basis of the process of software development, is not accurate.
2. Completeness : The software requirement specification should not be missing any of the requirements stated in the business requirements documentation that the user specified.
3. Prioritization of Requirements : Software Requirement Specification should not simply be a wish list. The requirements should follow the order of priority and preference.

3.1.3 Overview of responsibilities of Developer

The responsibilities from developer perspective are to develop the application as per the requirement with user friendly user interface. For this the developer is required to go through the concepts required to be implemented. The developer follows the software design life cycle, starting from requirement gathering, analysis, design, coding, and then implementation of the system. After the implementation is completed, the graphical data representation and memory utilization is tested manually for random files or data.

3.2 USAGE SCENARIO

To provide the analyzing data from different perspectives and summarizing it into useful information - information that can be used to increase revenue, cuts costs, or both. It can be useful where huge amount of data is to be store. Ex. Google, Social Networking Application Etc.

3.2.1 User profiles

Table 3.1: System Entities Description

Sr. No.	Actor	Description
1	Actor 1	Description 1
2	Actor 2	Description 2
3	Actor 3	Description 3

3.2.2 Use-cases

Mention the use cases in the tabular format.

Table 3.2: Use cases

Sr. No.	Use Case	Description	Actors	Assumption
1	Use case 1	Description 1	Actors	Assumption
2	Use case 2	Description 2	Actors	Assumption
3	Use case 3	Description 3	Actors	Assumption

3.2.3 Use Case View

A use case diagram is a graphical representation of a user's interaction with the system and depicting the specifications of a use case. A use case diagram can show the different types of users of a system and the various ways in which they interact with the system. Use case diagrams are used to gather the requirements of a system including internal and external influences. These requirements are mostly design requirements. So when a system is analyzed to gather its functionality use cases

are prepared and actors are identified. The purposes of use case diagrams can be as follows:

- Used to gather requirements of a system.
- Used to get an outside view of a system.
- Identify external and internal factors influencing the system.
- Show the interaction among the actors.

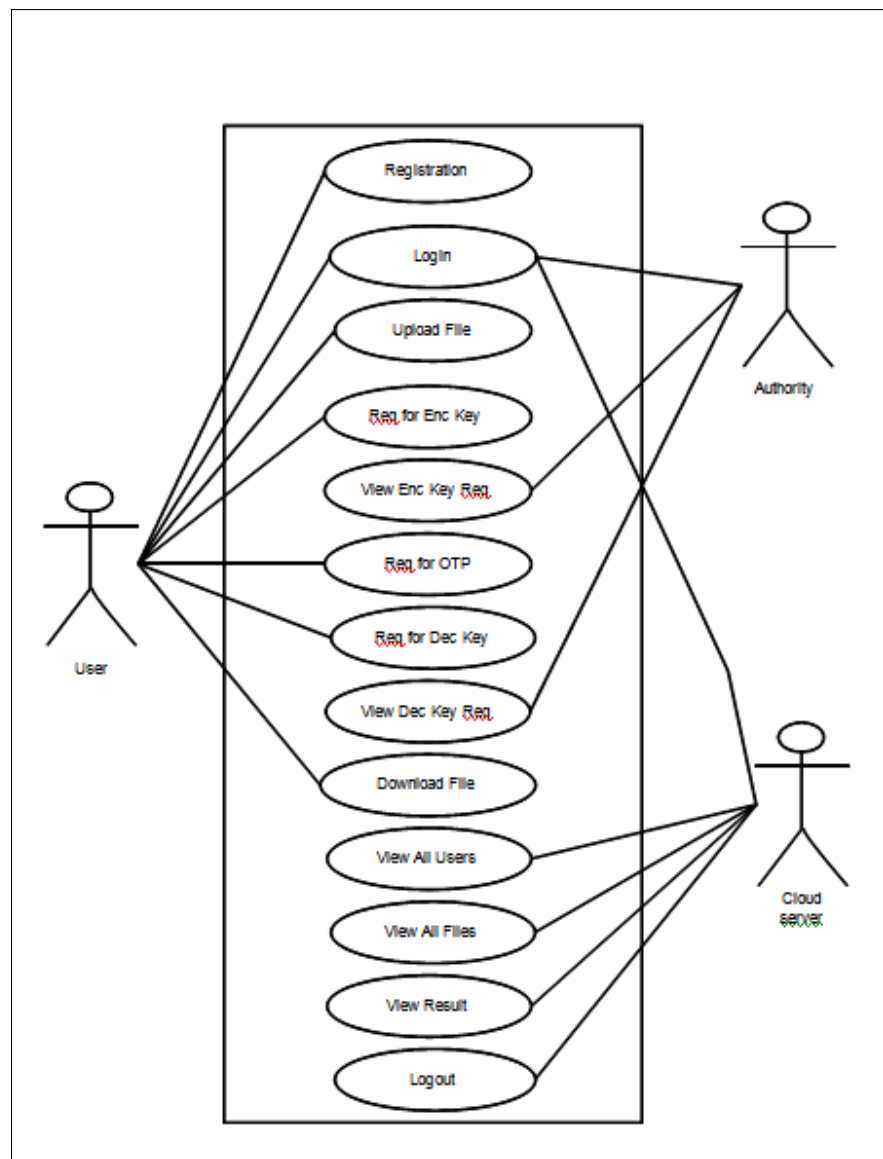


Figure 3.1: Use - Case diagram

3.3 DATA MODEL AND DESCRIPTION

3.3.1 Data Description

The controller is responsible for responding to user input and performs interactions on the data model objects. The controller receives the input; it validates the input and then performs the business operation that modifies the state of the data model.

3.3.2 Data objects and Relationships

The model is responsible for managing the data of the application. It responds to the request from the view and it also responds to instructions from the controller to update itself.

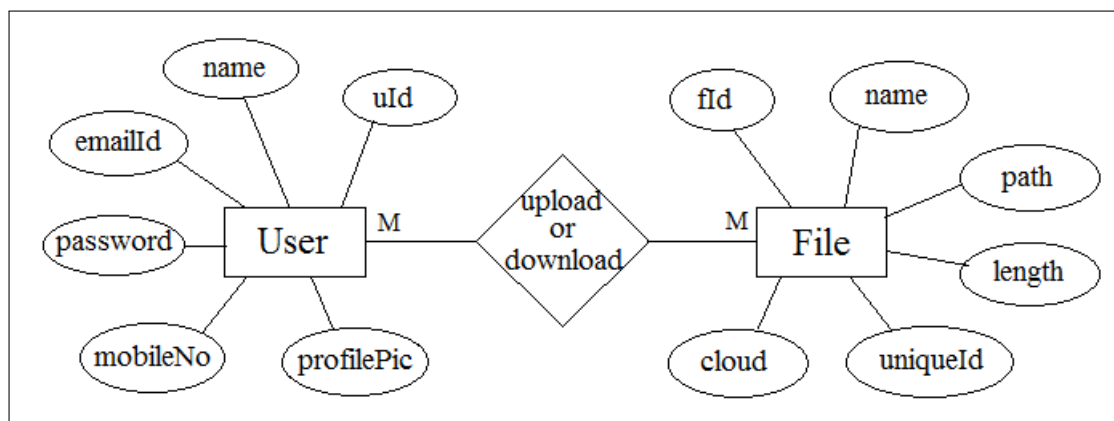


Figure 3.2: ER diagram

3.4 FUNCTIONAL MODEL AND DESCRIPTION

3.4.1 Data Flow Diagram

A data flow diagram (DFD) is a graphical representation of the “flow” of data through an information system, modeling its process aspects. Often they are a preliminary step used to create an overview of the system which can later be elaborated. DFDs can also be used for the visualization of data processing (structured design). A DFD shows what kinds of information will be input to and output from the system, where the data will come from and go to, and where the data will be stored. It does not show

information about the timing of processes, or information about whether processes will operate in sequence or in parallel (which is shown on a flowchart).

3.4.1.1 Level 0 Data Flow Diagram

The DFD Level 0 identifies external entities and processes of the system. Level 0 explains the architecture that would be used for developing a software product. A context diagram is a top level (also known as "Level 0") data flow diagram. It only contains one process node ("Process 0") that generalizes the function of the entire system in relationship to external entities.

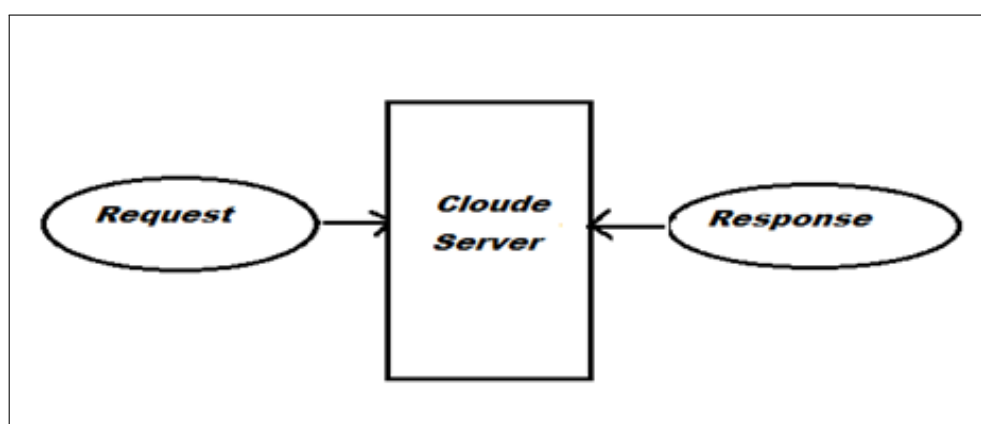


Figure 3.3: DFD0 diagram

3.4.1.2 Level 1 Data Flow Diagram

This DFD Level 1 shows the main processes in the work and the entities involved in it. We usually begin withdrawing a context diagram, a simple representation of the whole system. To elaborate further from that, we drill down to a level 1 diagram with additional information about the major functions of the system. Level 1 is an extension of level 0 diagram.

3.4.2 Description of functions

A description of each software function is presented. A processing narrative for function n is presented.(Steps)/ Activity Diagrams. For Example Refer 3.5

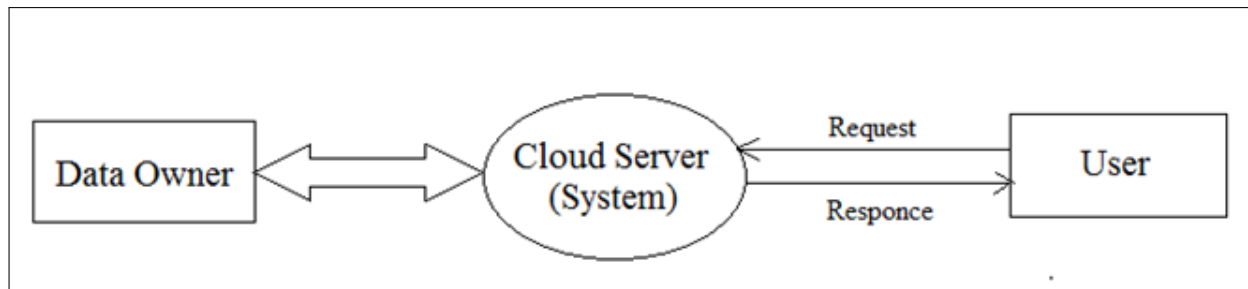


Figure 3.4: DFD1 diagram

3.4.3 Activity Diagram:

Activity diagrams are graphical representations of workflows of stepwise activities and actions with support for choice, iteration and concurrency. In the Unified Modeling Language, activity diagrams are intended to model both computational and organizational processes (i.e. workflows). Activity diagrams show the overall flow of control. Activity diagrams are constructed from a limited number of shapes, connected with arrows. The most important shape types:

- Rounded rectangles represent actions;
- Diamonds represent decisions;
- Bars represent the start (split) or end (join) of concurrent activities;
- A black circle represents the start (initial state) of the workflow;
- An encircled black circle represents the end (final state).

Arrows run from the start towards the end and represent the order in which activities happen. Hence they can be regarded as a form of flowchart. Typical flowchart techniques lack constructs for expressing concurrency. However, the join and split symbols in activity diagrams only resolve this for simple cases; the meaning of the model is not clear when they are arbitrarily combined with decisions or loops.

- Activity Diagram for User

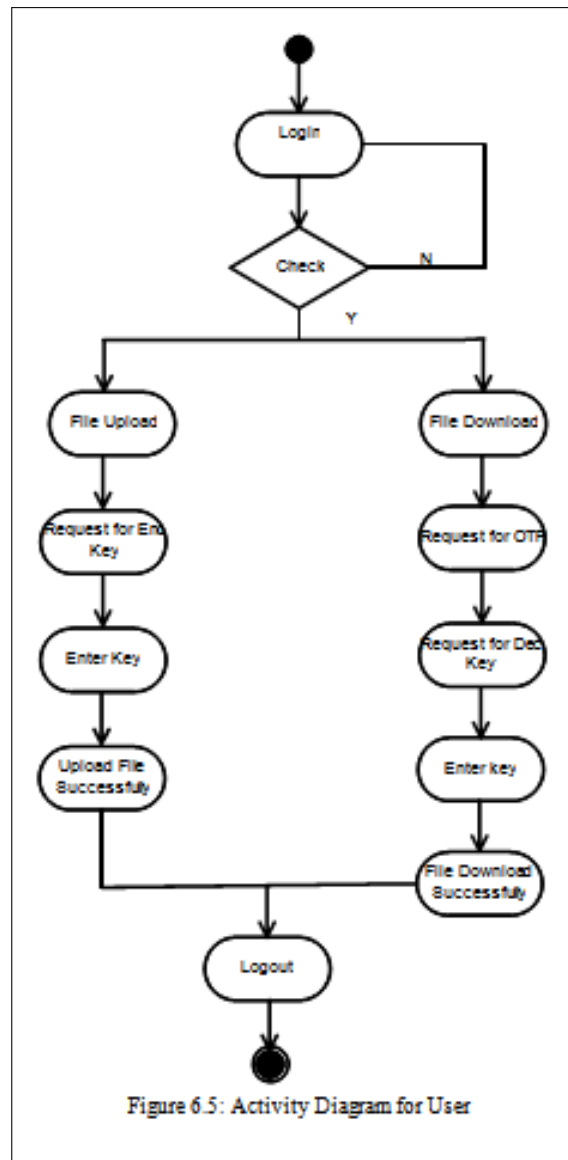


Figure 3.5: Activity diagram

3.4.4 Non Functional Requirements:

- Interface Requirements
- Performance Requirements
- Software quality attributes such as availability [related to Reliability], modifiability [includes portability, reusability, scalability] , performance, security, testability and usability[includes self adaptability and user adaptability]

3.4.5 State Diagram:

State Transition Diagram

Fig.3.6 example shows the state transition diagram of Cloud SDK. The states are represented in ovals and state of system gets changed when certain events occur. The transitions from one state to the other are represented by arrows. The Figure shows important states and events that occur while creating new project.

3.4.6 Design Constraints

Any design constraints that will impact the subsystem are noted.

3.4.7 Software Interface Description

The software interface(s) to the outside world is(are) described. The requirements for interfaces to other devices/systems/networks/human are stated.

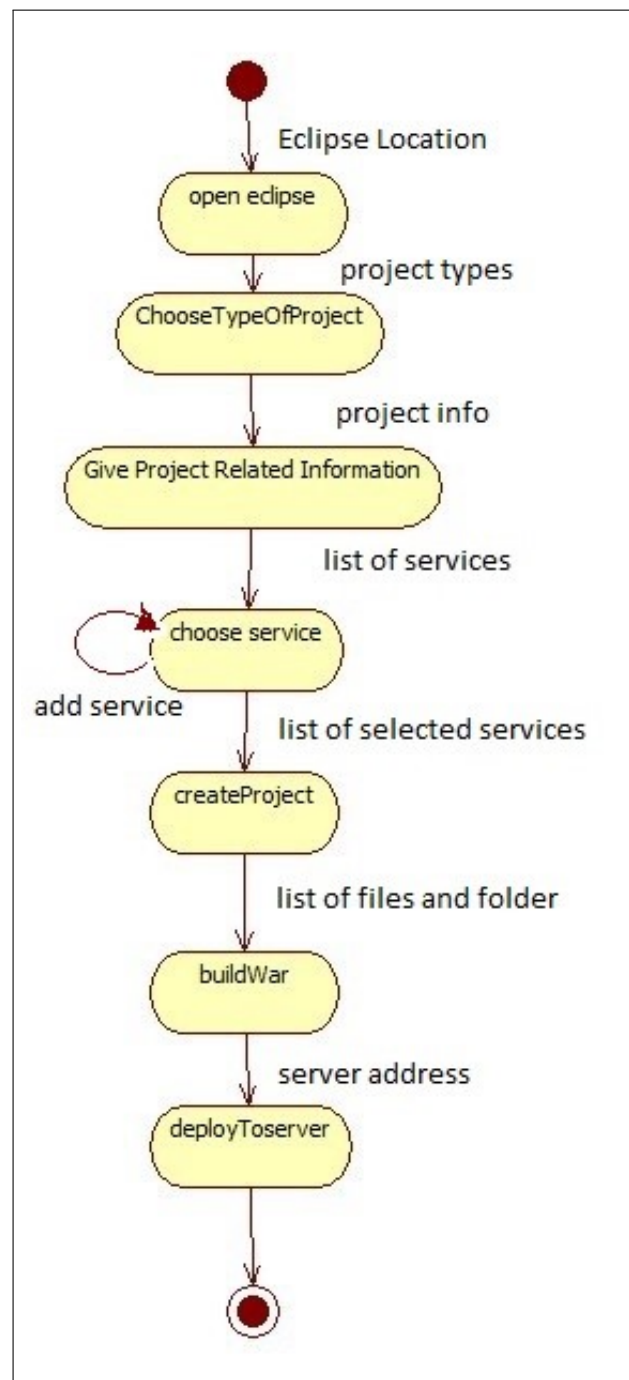


Figure 3.6: State transition diagram

CHAPTER 4

PROJECT PLAN

4.1 PROJECT ESTIMATES

Use Waterfall model and associated streams derived from assignments 1,2, 3, 4 and 5(Annex A and B) for estimation.

1. Requirement Gathering: Requirement gathering and plan for the initial part of the project was as follows: Understanding the problem definition. Understanding the reliable factors for data sharing systems. Gathering information about required software. Gathering information about required Hardware Resources Preparing preliminary design of overall work flows of project deciding the modules required for overall execution.
2. Analysis : Analysis of system collects systems requirement. Detailed plan of project and estimation of budget is studied in this stage. It is useful for the future development of project. Document containing system requirements is end product of this phase. Management and user requirements, alternative plans are further described by this document.
3. Design: Design focuses on high level design like, software i.e. Net beans for J2ME software are needed and their interaction with each other, then interface design to visualize how the project going to look like.
4. Implementation: In this phase code is generated from the designs.
5. Testing: System is tested in this phase. Individual modules are tested then whole system is tested. The system is tested to check inter related working of modules i.e. integration testing and that the system does what the user requires.
6. Maintenance: Maintenance is required for system. Because of some unexpected input values into the system the, changes in the system occurred.

4.1.1 Reconciled Estimates

4.1.1.1 Cost Estimate

4.1.1.2 Time Estimates

4.1.2 Project Resources

Project resources [People, Hardware, Software, Tools and other resources] based on Memory Sharing, IPC, and Concurrency derived using appendices to be referred.

4.2 RISK MANAGEMENT W.R.T. NP HARD ANALYSIS

This section discusses Project risks and the approach to managing them.

4.2.1 Risk Identification

For risks identification, review of scope document, requirements specifications and schedule is done. Answers to questionnaire revealed some risks. Each risk is categorized as per the categories mentioned in [1]. Please refer table 4.1 for all the risks. You can refer following risk identification questionnaire.

1. Have top software and customer managers formally committed to support the project?
2. Are end-users enthusiastically committed to the project and the system/product to be built?
3. Are requirements fully understood by the software engineering team and its customers?
4. Have customers been involved fully in the definition of requirements?
5. Do end-users have realistic expectations?
6. Does the software engineering team have the right mix of skills?
7. Are project requirements stable?
8. Is the number of people on the project team adequate to do the job?

9. Do all customer/user constituencies agree on the importance of the project and on the requirements for the system/product to be built?

4.2.2 Risk Analysis

The risks for the Project can be analyzed within the constraints of time and quality

Table 4.1: Risk Table

ID	Risk Description	Probability	Impact		
			Schedule	Quality	Overall
1	System Failure	Low	Low	High	High
2	Connection Failure	Low	Low	High	High

Table 4.2: Risk Probability definitions [1]

Probability	Value	Description
High	Probability of occurrence is	> 75%
Medium	Probability of occurrence is	26 – 75%
Low	Probability of occurrence is	< 25%

Table 4.3: Risk Impact definitions [1]

Impact	Value	Description
Very high	> 10%	Schedule impact or Unacceptable quality
High	5 – 10%	Schedule impact or Some parts of the project have low quality
Medium	< 5%	Schedule impact or Barely noticeable degradation in quality Low Impact on schedule or Quality can be incorporated

4.2.3 Overview of Risk Mitigation, Monitoring, Management

Following are the details for each risk.

Risk ID	1
Risk Description	Description 1
Category	Development Environment.
Source	Software requirement Specification document.
Probability	Low
Impact	High
Response	Mitigate
Strategy	Strategy
Risk Status	Occurred

Risk ID	2
Risk Description	Description 2
Category	Requirements
Source	Software Design Specification documentation review.
Probability	Low
Impact	High
Response	Mitigate
Strategy	Better testing will resolve this issue.
Risk Status	Identified

Risk ID	3
Risk Description	Description 3
Category	Technology
Source	This was identified during early development and testing.
Probability	Low
Impact	Very High
Response	Accept
Strategy	Example Running Service Registry behind proxy balancer
Risk Status	Identified

4.3 PROJECT SCHEDULE

4.3.1 Project task set

Major Tasks in the Project stages are:

- Task 1:Requirement Analysis (Base Paper Explanation).
- Task 2:Project Specification (Paper Work).
- Task 3:Technology Study and Design.
- Task 4:Coding and Implementation (Module Development).

4.3.2 Task network

Individual tasks and subtasks have interdependencies based on their sequence. A task network is a graphic representation of the task flow for a project. Project tasks and their dependencies are noted.

4.4 TEAM ORGANIZATION

The manner in which staff is organized and the mechanisms for reporting are noted.

4.4.1 Team structure

The team structure for the project is identified. Roles are defined.

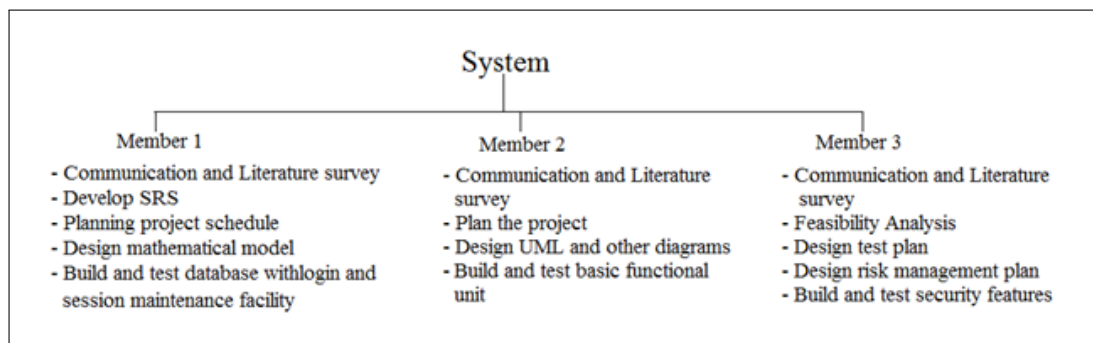


Figure 4.1: Team Structure

4.4.2 Management reporting and communication

Mechanisms for progress reporting and inter/intra team communication are identified as per assessment sheet and lab time table.

Table 4.4: Management plan [1]

Sr No.	Month	Description
1	June	Discussion with guide regarding domain.
		Searching for IEEE paper for domain.
2	July	Short listing of IEEE papers within domain.
		Selection of IEEE paper.
3	August	Deciding Project name.
		Submission of Synopsis.
4	September	Requirement analysis.
		Designing of models.
5	October	Report preparation.
		Stage-I report submission.

4.4.3 Timeline Chart

A project timeline chart is presented. This may include a time line for the entire project. Above points should be covered in Project Planner as Annex C and you can mention here Please refer Annex C for the planner

Table 4.5: Time-line Chart [1]

Task Name	Start date	End date	Duration
Initiate the project	04/8/2017	24/8/2017	21
Communication	04/8/2017	10/8/2017	7
Literature survey	11/8/2017	17/8/2017	7
Define scope	18/8/2017	19/8/2017	2
Develop SRS	20/8/2017	24/8/2017	5
Plan the project	25/8/2017	5/10/2017	42
Design mathematical model	25/8/2017	31/8/2017	7
Feasibility Analysis	01/9/2017	07/9/2017	7
Develop work breakdown structure	08/9/2017	09/9/2017	2
Planning project schedule	10/9/2017	14/9/2017	5
Design UML and other diagrams	15/9/2017	21/9/2017	7
Design test plan	22/9/2017	28/9/2017	7
Design risk management plan	29/9/2017	5/10/2017	7
Execute the project	05/01/2018	29/03/2018	84
Build and test basic functional unit	05/01/2018	25/01/2018	21
Build and test database with login	26/01/2018	15/02/2018	21
Build and test Bluetooth mode	16/02/2018	08/03/2018	21
Build and test security features	09/03/2018	29/03/2018	21

CHAPTER 5

SYSTEM DESIGN

5.1 INTRODUCTION

Brief introduction.

5.2 SYSTEM ARCHITECTURE DESIGN

Draw a system architecture diagram using advanced tool like draw.io , canva etc.
Dont use AI generated diagrams as it is. Explain the architecture in detail.

5.3 MATHEMATICAL MODEL FOR THE PROPOSED SYSTEM

Include detailed mathematical model for the proposed system model.

5.4 DATA DESIGN

May include data design part. Internal software data structure

Global data structure

Temporary data structure

Database description

COMPOENT DESIGN

5.5 RESULTS AND DISCUSSION

CHAPTER 6

OTHER SPECIFICATIONS

6.1 ADVANTAGES OF THE SYSTEM

Explain advantages in detail.

6.2 LIMITATIONS OF THE SYSTEM

Explain limitations in detail.

6.3 APPLICATIONS OF THE SYSTEM

Explain applications in detail.

CHAPTER 7

VALIDATION AND TESTING

7.1 TESTING METHODOLOGY

Explain testing methodology by using suitable diagram and brief explanation.

7.2 TEST CASES

Brief summary of test cases and insert following table.

Table 7.1: Test Case Summary

Sr. No	Test Case Title / Description	Testing Methodology Applied	Result (PASS/FAIL)	Defects Identified
1	Login functionality with valid credentials	Black box testing	PASS	None
2	Invalid password handling	Boundary value analysis	FAIL	Error message not displayed properly
3	Session timeout validation	System testing	PASS	None

CHAPTER 8

CONCLUSION FUTURE SCOPE

8.1 CONCLUSION

Write one page conclusion and future scope

8.2 FUTURE SCOPE

CHAPTER 9

REFERENCES

- [1] R. S. Pressman, *Software Engineering (3rd Ed.): A Practitioner's Approach*.
New York, NY, USA: McGraw-Hill, Inc., 1992.

ANNEXURE A

PLAGIARISM REPORT

Add only first page of the plagiarism report of your project received from Project Coordinator. (Do not add complete plagiarism report.)

ANNEXURE B

COPYRIGHT

Add the registered copyright certificate and copyrighted document.

If the copyright is in process, then add the acknowledgement receipt along with the copyrighted document.

ANNEXURE C

PATENTS

Add the Design/ Utility Patent related documents.

ANNEXURE D

RESEARCH PUBLICATION

Add the Journal/Conference research papers here. Make sure to add complete research paper from first page till the last page. If you have multiple publications, add all of them here.

Add published paper. Don't add unpublished paper draft.

ANNEXURE E

CONFERENCE CERTIFICATES

Add the conference participation certificates here for all the group members and project guide.

ANNEXURE F

PROJECT COMPETITION

PARTICIPATION

Add Project Competition participation certificates.

You can add any other achievements related to BE Project here.

ANNEXURE G

SPONSORSHIP DOCUMENTS

Attach sponsorship documents if the project is a sponsored project.